

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING / AIML  
ABES Engineering College / AKTU, Ghaziabad

# SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title: CardioPulse AI-- Heart Disease Risk Prediction

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# 1. Introduction

## 1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a complete, clear, and unambiguous definition of the functionality, external interfaces, performance targets, and constraints of the proposed system: 'CardioPulse AI — Heart Disease Risk Prediction & Animated Diagnostic System'. This specification serves as a binding reference agreement between student developers, project evaluators, academic supervisors, and prospective end users during the 3rd-year engineering curriculum.

## 1.2 Scope of the System

The proposed application is designed to solve the critical need for rapid, non-invasive cardiovascular risk assessment by integrating K-Nearest Neighbors (KNN) Machine Learning classification with a dynamic, animated Streamlit UI, physical moving needle risk meter, ReportLab PDF report generation, and cartoon health action initiatives by delivering an end-to-end automated platform. Key boundary guidelines include:

- **In-Scope Capabilities:** Interactive 11-feature patient physiological data entry, standardized feature scaling ('heart\_scaler.pkl'), KNN classification ('knn\_heart\_model.pkl'), Plotly physical moving needle risk meter, ReportLab PDF diagnostic report download, session history log tracking (CSV export), and cartoon Dr. Hearty health action initiative cards.
- **Out-of-Scope Boundaries:** Invasive surgical monitoring hardware fabrication, real-time hospital EHR database integration, and live commercial telemedicine video consultations.
- **Expected Benefits:** Immediate heart disease risk estimation in under 1.0 second, enhanced patient engagement through animated Dr. Hearty cartoon mascot visuals, and instant downloadable PDF clinical summary reports.

## 1.3 Definitions, Acronyms, and Abbreviations

|                  |                           |                                                                         |
|------------------|---------------------------|-------------------------------------------------------------------------|
| Standard         | <b>SRS</b>                | Software Requirements Specification (IEEE Std 830-1998 format).         |
| Architecture     | <b>API</b>                | Application Programming Interface for data interchange via JSON/HTTPS.  |
| Security         | <b>JWT / RBAC</b>         | JSON Web Token for stateless authorization; Role-Based Access Control.  |
| Database         | <b>CRUD</b>               | Create, Read, Update, Delete basic data manipulation operations.        |
| Network          | <b>REST</b>               | Representational State Transfer stateless software architectural style. |
| Project Specific | <b>&lt;Add Custom&gt;</b> | e.g., YOLOv8 (Object Detection) / BLE (Bluetooth Low Energy) / MQTT.    |
| Machine Learning | <b>KNN</b>                | K-Nearest Neighbors classification algorithm for heart risk prediction. |

|          |     |                                                                       |
|----------|-----|-----------------------------------------------------------------------|
| Medical  | ECG | Electrocardiogram measuring electrical activity of the cardiac cycle. |
| Medical  | BP  | Resting Blood Pressure measured in mm Hg.                             |
| Document | PDF | Portable Document Format generated via ReportLab library.             |

## 1.4 References

- **IEEE Std 830-1998:** IEEE Recommended Practice for Software Requirements Specifications.
- **Software Engineering (10th Edition):** Ian Sommerville, Pearson Education.
- Relevant Research Papers & APIs: Heart Disease UCI Dataset (Kaggle/UCI Machine Learning Repository); Scikit-Learn (v1.3+) Documentation; Streamlit Web Framework Documentation; ReportLab PDF Library Manual..

## 1.5 Document Overview

The remainder of this SRS is partitioned into structural sections: Section 2 outlines the product perspective, target user personas, operational context, and design constraints. Section 3 itemizes detailed functional requirements organized into logical software modules. Section 4 specifies external interfaces (UI, Hardware, Software, Network). Section 5 defines non-functional attributes (Performance, Reliability, Security). Finally, Section 6 provides standard academic modeling diagrams.

## 2. Overall Description

### 2.1 Product Perspective

The project is structured as a self-contained multi-tier system (comprising a presentation layer, an application/API logic tier, and a persistent database backend). It interfaces with external services (such as cloud storage or messaging gateways) strictly through secure, authenticated RESTful endpoints.

### 2.2 User Classes and Characteristics

The target software accommodates distinct user personas with varying access hierarchies:

|                        |                          |                                                                                                                                                   |
|------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| Patient / General User | Basic (Web Browser User) | Inputs vitals (Age, BP, Cholesterol, ECG, Max HR), views real-time moving needle risk meter, downloads PDF/TXT report, tracks assessment history. |
| Clinician / Doctor     | Medical Professional     | Reviews automated KNN risk predictions, inspects downloadable PDF assessment reports, evaluates patient action initiatives.                       |
| Guest / Public Visitor | Basic                    | Read-only access to promotional landing pages, system overview, public catalogues, and self-registration portal.                                  |

### 2.3 Operating Environment

The application operates within the following software and hardware environments:

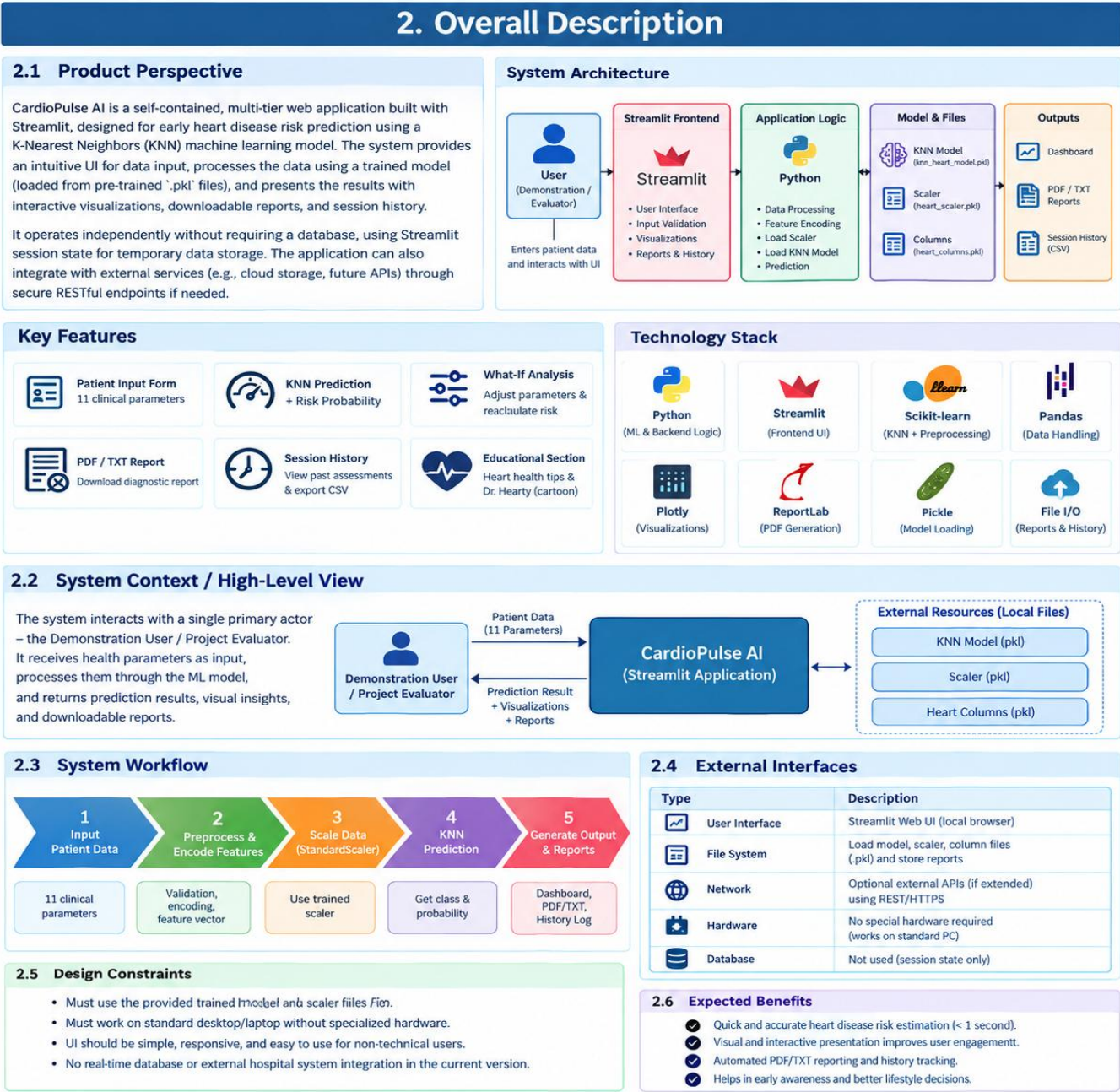
- **Client Requirements:** Modern web browser (Google Chrome >= v110, Mozilla Firefox >= v108, Edge, Safari); minimum display resolution 1366x768 for desktop, responsive view down to 360x640.
- **Server Runtime Environment:** Linux (Ubuntu 22.04 LTS / Debian 12) or Windows Server; Python 3.10+ / Node.js 18 LTS / Java 17 LTS.
- **Database Storage Tier:** PostgreSQL 15 / MySQL 8.0 / MongoDB 6.0 Enterprise or community editions.

### 2.4 Design and Implementation Constraints

- **Academic Budget Constraint:** Utilization of free-tier cloud resources (e.g., Render, Railway, AWS Educate, or local containerized hosting).
- **Security Compliance:** Passwords stored exclusively as salted hashes (PBKDF2 or Bcrypt); token invalidation upon user logout.
- **Modularity Standards:** Adherence to Model-View-Controller (MVC) or modular REST service patterns to ensure clean separation of concerns.

2.5 Assumptions and Dependencies

- **Assumptions:** Users possess continuous Internet connectivity with minimal bandwidth of 2 Mbps; client clocks are synchronized via NTP.
- **Dependencies:** Continuous availability of third-party public APIs (e.g., Google OAuth 2.0, Geocoding service, or external model repository).



### 3. System Features and Functional Requirements

Requirements are categorized by functional subsystem modules with unique requirement identifiers (FR-x.x):

#### 3.1 Module 1: Authentication and Access Governance

|        |                              |                                                                                                                                        |        |
|--------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|--------|
| FR-1.1 | Physiological Data Ingestion | Captures 11 patient features (Age 18-100, BP 50-250, Cholesterol 50-700, MaxHR 60-250, ST Depression, Chest Pain, ECG, Angina, Slope). | High   |
| FR-1.2 | Preset Clinical Profiles     | Provides sample patient presets (Normal Low Risk, Asymptomatic High Risk, Borderline Stress) for instant input population.             | Medium |
| FR-1.3 | Session Token Management     | Generates signed JWT (HS256) valid for 120 minutes                                                                                     | Medium |

#### 3.2 Module 2: Core Processing & Business Operations

|        |                                         |                                                                                                                                      |      |
|--------|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------|
| FR-2.1 | Standard Scaling & KNN Inference Engine | Transforms 11 features using `heart_scaler.pkl` and predicts binary cardiovascular risk using `knn_heart_model.pkl`.                 | High |
| FR-2.2 | Moving Needle Risk Meter & Animations   | Calculates trigonometric needle pointer angle from probability score and renders animated Plotly radial gauge meter & snow/balloons. | High |
| FR-2.3 | Persistent Data Records Storage         | Writing processed states with relational foreign-key integrity                                                                       | High |

### 3.3 Module 3: Reporting, Visualization & Administrative Control

|        |                                             |                                                                                                                             |      |
|--------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|------|
| FR-3.1 | Dr. Hearty Recommended Action Initiatives   | Renders interactive cartoon cards for 150-min cardio power, heart-smart diet, stress control, and routine medical checkups. | High |
| FR-3.2 | PDF/TXT Report Export & Session History Log | Generates downloadable ReportLab PDF & TXT diagnostic reports and logs session runs with full CSV history export.           | High |
| FR-3.3 | User Status Moderation (Admin)              | Toggle active/disabled user status; manage platform parameters                                                              | High |

## 4. External Interface Requirements

### 4.1 User Interfaces (UI)

The interface will follow standard UX usability best practices (clean navigation bar, breadcrumbs, modal validation dialogs). All forms must provide immediate visual feedback (success banners or inline error alerts under erroneous fields).

[User Interface Layout & Screens]

1. Input Section: Patient Vitals & Cardiac Stress Forms (11 Clinical Features)
2. Diagnostic Output: Moving Needle Risk Meter, High/Low Risk Banners, PDF Download Button
3. Action Initiatives: Cartoon Dr. Hearty Patient Action Cards (Cardio, Diet, Stress, Checkups)

### 4.2 Hardware Interfaces

Specification of physical interaction points:

- **Standard Deployment:** Standard input/output peripherals (Keyboard, Mouse, Display  $\geq$  720p).
- **Embedded / IoT Variants (if applicable):** Microcontroller I/O interfaces (e.g., ESP32 GPIO, I2C, SPI sensors, USB UART serial communications).

### 4.3 Software Interfaces

- **Relational Database Connector:** SQLAlchemy / Sequelize ORM interfacing with MySQL/PostgreSQL via standard TCP ports.

- **External Cloud Utilities:** Amazon Web Services (S3 bucket for document storage) / Cloudinary for image optimization.

#### 4.4 Communication Interfaces

- **Web Traffic Protocol:** HTTPS (Hypertext Transfer Protocol Secure over Port 443 with TLS 1.3).
- **Data Serialization:** All payload bodies exchanged using standard UTF-8 encoded application/json format.

### 5. Non-Functional Requirements (NFRs)

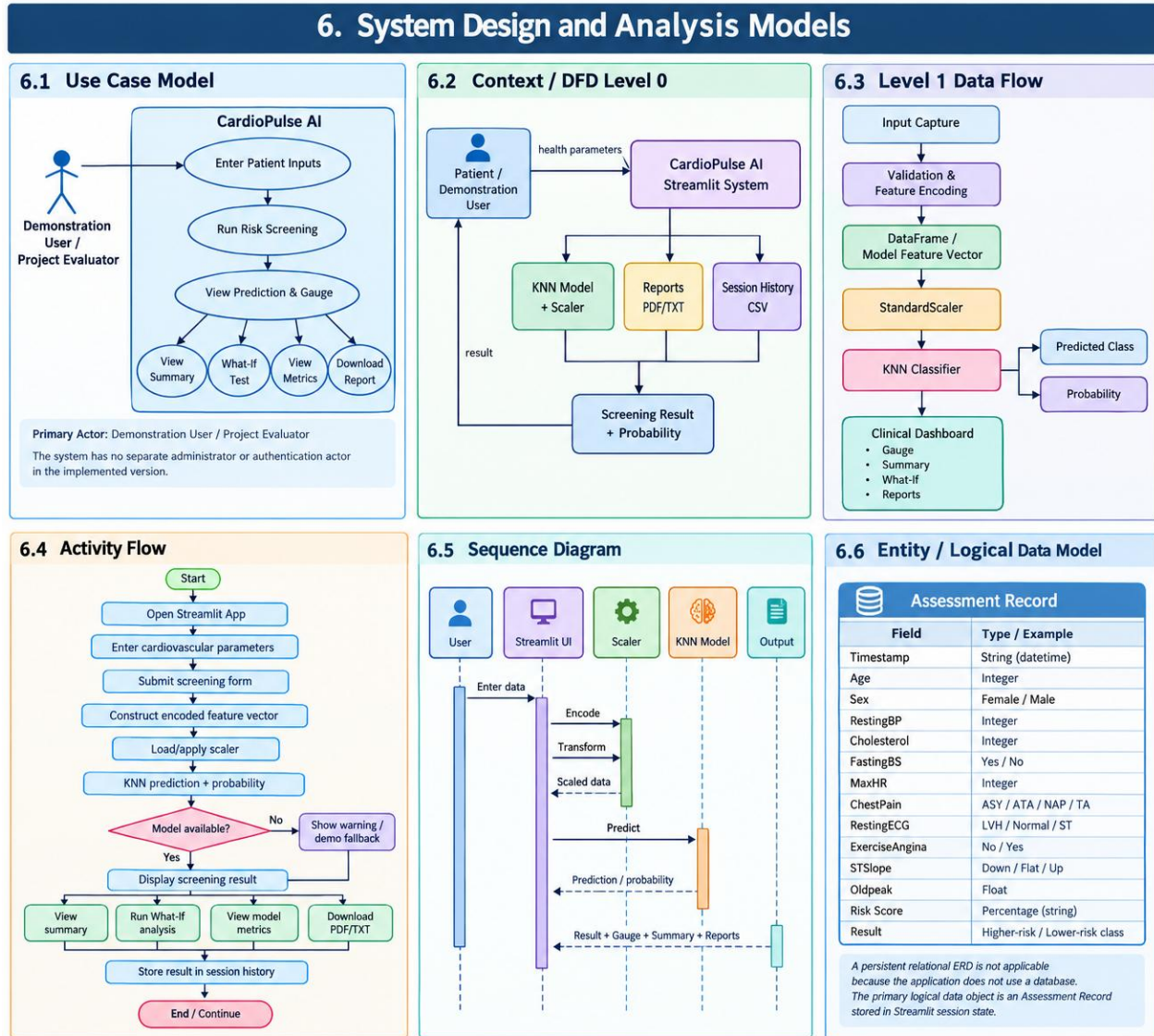
|                           |                                                                                                       |                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| 5.1 Performance & Latency | ML inference latency < 100ms;<br>Streamlit UI refresh < 500ms;<br>PDF report generation < 1.0 second. | Empirical benchmarking using Python time module & Streamlit runtime.  |
| 5.2 Usability & Security  | Interactive Dr. Hearty mascot visuals, dark glassmorphism theme; local session-state data isolation.  | Heuristic usability testing & client-side session isolation review.   |
| 5.3 Reliability           | System uptime >= 98% during academic review; automatic session recovery on client disconnect.         | Stress testing & server process monitor (PM2 / systemd auto-restart). |
| 5.4 Portability           | Cross-browser compatibility (Chrome, Edge, Firefox, Safari); responsive viewport (mobile to desktop). | Responsive viewport inspection in Developer Tools and BrowserStack.   |

## 6. System Design and Analysis Models

University evaluation rubrics require standard graphical analysis models illustrating the dynamic and static behavior of the application:

The following graphical models describe the implemented CardioPulse AI Streamlit workflow, including user interaction, data flow, KNN inference, reporting, session history, and the logical Assessment Record.

**Figure 6.1 — System Design and Analysis Models**



### Models included:

- 6.1 Use Case Model;
- 6.2 Context / DFD Level 0;
- 6.3 Level 1 Data Flow;
- 6.4 Activity Flow;

6.5 Sequence Diagram;

6.6 Entity / Logical Data Model.

## 7. Academic Review & Sign-Off

This Software Requirements Specification document has been submitted and reviewed by the academic supervisory panel:

Signature: \_\_\_\_\_  
Date: \_\_\_\_\_

Signature: \_\_\_\_\_  
Date: \_\_\_\_\_

Signature: \_\_\_\_\_  
Date: \_\_\_\_\_